

PAJU, Korea, Republic of

WMO: 470990

Lat: 37.8858N

Lon: 126.7664E

Elev: 31

StdP: 100.95

Time Zone: 9.00 (E09)

Period: 03-19

WBAN: 99999

Annual Heating, Humidification and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-16.1	-13.8	-23.0	0.5	-10.6	-21.0	0.6	-9.6	5.8	-3.7	5.3	-3.8	0.9	140	0.393

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	7.8	32.0	25.3	30.6	24.2	29.3	23.3	26.6	30.1	25.8	28.8	25.1	27.9	2.1	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
25.6	21.0	28.4	25.0	20.1	27.6	24.3	19.3	26.8	83.4	30.4	80.2	28.8	77.2	28.1	29.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
5.5	4.9	4.3	-18.9	34.1	3.3	0.8	-21.3	34.7	-23.3	35.2	-25.1	35.7	-27.6	36.3		
			DB	WB	DB	WB	DB	WB	DB	WB	DB	WB	DB	WB		
			-19.4	27.4	3.2	1.2	-21.7	28.2	-23.5	29.0	-25.3	29.7	-27.6	30.6		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	11.4	-4.6	-1.4	4.4	10.9	17.1	21.8	24.7	25.4	20.5	13.6	5.8	-2.3
	DBStd	10.92	4.51	4.33	3.70	3.36	2.66	2.08	2.04	2.53	2.55	3.68	4.60	4.52
	HDD10.0	1507	451	319	177	29	0	0	0	0	0	12	139	381
	HDD18.3	3147	709	552	431	224	56	1	0	0	9	150	376	639
	CDD10.0	2016	0	0	4	55	221	354	454	477	314	123	14	0
	CDD18.3	616	0	0	0	1	18	106	196	219	73	4	0	0
	CDH23.3	4365	0	0	0	13	176	657	1329	1804	367	19	0	0
	CDH26.7	1183	0	0	0	2	23	146	358	607	47	1	0	0
Wind	WSAvg	1.6	1.3	1.6	2.0	2.0	1.8	1.5	1.6	1.5	1.4	1.3	1.5	1.4
Precipitation	PrecAvg	1411	16	28	42	95	98	135	460	269	147	50	48	22
	PrecMax	2134	52	71	92	174	163	378	1102	606	466	183	110	63
	PrecMin	706	2	1	13	37	19	43	177	38	30	11	7	4
	PrecStd	355	13	21	25	34	44	83	239	178	122	42	29	14
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	8.0	13.1	18.6	25.0	28.9	31.5	32.9	34.1	29.4	25.5	19.7	10.4
		MCWB	4.1	7.8	10.4	14.3	17.0	21.2	25.3	26.5	22.0	17.8	14.2	6.4
	2%	DB	5.4	10.0	15.5	21.9	27.0	29.6	31.0	32.7	28.0	23.3	17.1	8.2
		MCWB	1.8	4.8	9.0	12.4	16.8	20.2	24.6	26.1	21.0	16.5	12.7	5.3
	5%	DB	3.8	7.6	13.3	19.9	25.2	28.3	29.9	31.5	26.9	21.6	15.1	6.3
		MCWB	0.7	3.8	7.6	11.5	16.2	20.2	24.1	25.5	20.3	15.6	11.0	3.5
	10%	DB	2.4	5.5	11.4	17.9	23.7	27.1	28.8	30.0	25.6	19.9	13.2	4.5
		MCWB	-0.1	2.0	6.4	10.9	15.7	19.9	23.7	24.9	19.7	14.5	9.9	2.1
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	4.5	9.7	12.4	16.8	20.0	23.6	27.0	28.0	24.3	19.9	15.8	8.1
		MCDB	7.2	11.9	16.0	23.1	24.0	27.4	30.9	31.8	26.7	22.9	17.7	9.5
	2%	WB	2.6	5.7	10.5	14.5	18.7	22.7	26.1	27.1	23.1	18.0	13.9	5.6
		MCDB	4.6	9.4	14.0	18.1	23.2	26.5	29.1	30.9	25.7	21.3	15.9	7.8
	5%	WB	1.3	4.0	8.7	13.1	17.8	21.9	25.4	26.4	22.1	16.8	12.0	3.8
		MCDB	3.3	7.1	12.3	17.4	22.5	25.5	28.1	29.7	24.5	20.1	14.5	5.9
	10%	WB	0.0	2.4	7.0	11.8	16.8	21.1	24.8	25.7	21.2	15.5	10.3	2.4
		MCDB	2.2	5.1	10.6	16.4	21.8	24.8	27.2	28.7	24.0	18.8	12.7	4.3
Mean Daily Temperature Range	5% DB	MDBR	11.0	11.0	11.4	12.2	11.8	9.6	6.8	7.8	9.7	12.1	10.5	9.7
		MCDBR	11.2	13.1	14.0	15.4	14.8	12.1	9.1	9.4	10.9	12.8	12.1	10.5
		MCWBR	8.1	8.6	8.0	7.2	5.9	4.3	3.6	3.6	4.6	6.1	7.5	7.8
	5% WB	MCDBR	9.7	11.8	11.7	12.1	11.2	8.6	6.8	8.0	8.1	10.7	10.2	9.9
		MCWBR	7.5	8.5	7.9	6.7	5.3	3.8	3.1	3.6	4.2	5.8	7.4	7.9
Clear-Sky Solar Irradiance	taub	0.425	0.485	0.564	0.616	0.627	0.700	0.672	0.578	0.508	0.503	0.481	0.426	
	taud	1.934	1.837	1.712	1.650	1.676	1.596	1.704	1.920	2.029	1.917	1.913	1.962	
	Ebn at Noon	703	711	697	692	696	646	660	715	742	687	638	666	
	Edh at Noon	139	175	218	246	244	264	235	186	157	159	142	127	
All-Sky Solar Radiation	RadAvg	2.50	3.26	4.26	4.96	5.58	5.22	4.05	4.34	4.17	3.57	2.42	2.18	
	RadStd	0.16	0.25	0.41	0.35	0.54	0.55	0.61	0.60	0.44	0.24	0.24	0.16	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (43 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+104	+57

Nomenclature: See separate page